Building and Environment 136 (2018) 185–197

Contents lists available at ScienceDirect

Building and Environment

journal homepage: www.elsevier.com/locate/buildenv

Indoor air quality in classrooms: Environmental measures and effective ventilation rate modeling in urban elementary schools

David L. Johnson[a,*], Robert A. Lynch[b], Evan L. Floyd[a], Jun Wang[a], Jacob N. Bartels[a]

[a] Department of Occupational and Environmental Health, University of Oklahoma College of Public Health, P.O. Box 26901, Oklahoma City, OK 72126-0901, United States

[b] Xi'an Jiaotong - Liverpool University, 111 Ren'ai Road, Suzhou Dushu Lake Higher Education Town, Jiangsu Province, 215123, China

ARTICLE INFO

Keywords:
Ventilation
Schools
Indoor air quality
Transient mass balance

ABSTRACT

Associations have been shown between poor classroom indoor air quality (IAQ) and schoolchildren's risk of asthma, increased absenteeism, and impaired performance on standardized tests. Mechanically ventilated classrooms often lack an adequate fresh air supply. There is also concern that outdoor pollutants, particularly vehicle exhaust products, may penetrate classrooms. The purpose of this work was to characterize IAQ in elementary school classrooms and estimate average effective fresh air ventilation rates under cold, mild, and warm season conditions. IAQ measures were made in third-grade classrooms of 12 elementary schools. Particulate matter, CO_2, CO, NO_2, total VOCs, and formaldehyde concentrations, as well as relative humidity and temperature, were measured for 24-h periods in each season. Effective fresh air ventilation rates were estimated using a transient mass balance modeling approach. The schools measured had generally adequate temperature and humidity control, extremely low non-occupant related pollutants, and little to no incursion of outdoor vehicle-related pollutants. However, there was a lack of adequate fresh air ventilation in many cases. Ventilation adequacy varied within the schools across seasons but with no consistent pattern, perhaps reflecting variations in class size as well as seasonal demands on the HVAC systems and/or HVAC seasonal operating mode. Transient mass balance method effective fresh air ventilation estimates near or above ASHRAE-recommended fresh air ventilation rates for people-related pollutants corresponded well with good CO_2 control in the classrooms.

1. Background

Poor indoor air quality (IAQ) has long been recognized as a cause of occupant discomfort, adverse health effects, increased absenteeism from work or school, and degraded cognitive performance [1]. Some contaminants may produce a building-related illness that can be objectively verified and attributed to a specific cause, e.g. elevated carboxyhemoglobin associated with carbon monoxide from a back-drafting combustion appliance. However, the more common manifestation or poor IAQ is via nonspecific symptoms such as headache, eye or nasal irritation, skin rash or itch, malaise, or difficulty concentrating. Such symptoms cannot usually be attributed to a specific cause and their occurrence is often described as "sick building syndrome" or "tight building syndrome" [2]. Indoor contaminants contributing to such symptoms may include particulate matter (PM), volatile organic compounds (VOCs) including aldehydes (especially formaldehyde), and oxidative gases such as nitrogen dioxide (NO_2) and ozone. These contaminants may be emitted by office equipment, released by cleaning and disinfection solutions or other consumer products, or off-gassed from building materials, paints, and furnishings. Pollutants may also enter the indoor environment from outdoors. For example, vehicle exhaust components including carbon monoxide (CO), nitrogen oxides (NO_x), and particulate matter can make their way in via infiltration through openings or leaks in the building envelope or via air drawn in by the heating, ventilating, and air conditioning (HVAC) system. Finally, people-related indoor pollutants, contributed primarily by expired air, can play a major if not dominant role in poor IAQ. Pollutants in exhaled air include carbon dioxide (CO_2) in percent concentrations (approximately 4000–5000 parts per million [ppm]), numerous VOCs typically in part per billion (ppb) concentrations, and water vapor.

CO_2 concentration is routinely used as a sentinel indicator of people-related indoor pollutants, including those perceived as "body odor". As discussed in the ASTM Standard Guide for Using Indoor Carbon Dioxide Concentrations to Evaluate Indoor Air Quality and Ventilation (ASTM D6245) [3], research has shown that body odors in a space will not be offensive to most entering persons if people-related CO_2 is maintained at less than about 650 ppm above background levels. Of course, it is not the CO_2 that is offending, but rather the VOCs and

* Corresponding author. University of Oklahoma Department of Occupational and Environmental Health, 801 NE 13th Street, Room 413, Oklahoma City, OK 73104, United States.
E-mail address: David-Johnson@ouhsc.edu (D.L. Johnson).

https://doi.org/10.1016/j.buildenv.2018.03.040
Received 15 February 2018; Received in revised form 23 March 2018; Accepted 24 March 2018
Available online 28 March 2018
0360-1323/

other gaseous compounds in exhaled breath. CO_2 has been used as an indicator because it is easily measured and its concentration is correlated with the concentration of the offending species. The familiar fresh air ventilation guideline that CO_2 in indoor spaces should be maintained below 1000 ppm to control people-related pollutants resulted from adding the 650 ppm value to the normal background CO_2 concentration in urban air, which is about 300–350 ppm [3]. Thus, CO_2 was not viewed as a pollutant of concern, but rather as an indicator of how well other people-related pollutants are controlled – particularly odor-causing compounds. However, recent controlled studies have shown that CO_2 alone can cause moderate decrements in decision-making performance in adults at concentrations as low as 1000 ppm, with large decrements above 2500 ppm [4,5]. No similar studies have been reported for its effects on children.

Children are a subpopulation of special interest for many environmental exposures, and the effects of poor school IAQ on schoolchildren's health and academic performance is an active area of research. Associations have been shown between poor IAQ and adverse health effects including asthma, absenteeism, and impaired performance on standardized tests [6–13]. A number of studies have shown that mechanically ventilated classrooms often lack an adequate fresh air supply in schools in the US [6,11,13,14] as well as in England [15–19] and Europe [7,20–23].

The purpose of this work was to characterize indoor air quality in elementary school classrooms and estimate the average effective fresh air ventilation rate of their HVAC systems under cold, mild, and warm season conditions using a transient mass balance modeling approach.

2. Methods

2.1. IAQ measurements

Indoor air quality measures were conducted in 12 elementary schools in the Oklahoma City metropolitan area. We elected to conduct air sampling in third grade classrooms only. In the schools studied, students in that grade spent the majority of their day in a single classroom, so that the occupancy and gender distribution were known. Continuous measurement of IAQ indicators was performed over 24-h periods. The schools are designated A through L as shown in Table 1. Their original build dates ranged from 1919 to 2011, and all of the schools were built or received some level of renovation under the city's Metropolitan Area Projects Plan for Kids (MAPS for Kids) school enhancement program since 2000. The subject schools were chosen to represent the range of renovation from minimal work on an existing structure (schools C and F) through classroom additions and core facility expansions (schools B, E, and G through L) to all-new construction (schools A and D). The measurements were conducted during February–March, April–May, and August–September 2016 to represent cold, mild, and warm season conditions, respectively, though only four of the schools could be measured under all three conditions. The number of seasons sampled per school is shown in Table 1. A total of 33 days of sampling was conducted in 12 schools, with 58 sets of measures taken. Ten of the 12 schools were sampled three times, one school was sampled only once, and another only twice. Four of the schools were sampled in all three seasons. An instrument set was placed outdoors on 4 occasions (3 in February and 1 in May), and in a second classroom on 24 occasions.

Measurement instruments were placed along one of the room walls to avoid interfering with instruction and to minimize the potential for curious students to interfere with the measurements. Both direct-reading and sorbent tube samplers were employed. Direct reading instruments included: (1) a multi-sensor IAQ meter (WolfPack Area Monitor, GrayWolf Sensing Solutions, Shelton, CT) for CO_2, CO, NO_2, total VOCs by photoionization, temperature, and relative humidity, with 1-min time resolution; (2) a single-sensor colorimetric coupon formaldehyde detector with 30-min resolution (GrayWolf FM-801 Formaldehyde Multimode Monitor); and (3) a 15-channel aerosol spectrometer with 0.3–20 μm (μm) particle size detection and 6-s time resolution (Model 1.108 Portable Aerosol Spectrometer, Grimm Tecknik GmBH, Ainring, Germany). The IAQ meter sensors were periodically calibrated with zero air and span gases. The instruments ran continuously for 24 h. In addition, sequential 12-h trisorbent thermal desorption tube samples were collected at 100 mL per minute. One instrument of each type and three sorbent tubes (in a cluster) were placed in two classrooms or in one classroom and outdoors. When only one classroom was fully instrumented, additional single sorbent tubes were placed in two nearby third grade classrooms to provide a basis (VOCs) for comparison. Outdoor sampling could not be performed on all occasions due to either weather conditions or equipment security issues. Room dimensions and student headcount by gender were recorded for each sampling event. The same classroom was usually measured in all visits. It was not possible to assess the HVAC systems' characteristics or their operating modes. After sampling, the logged data were downloaded to a computer for analysis, and sorbent tubes were refrigerated at 4 °C for later gas chromatography – mass spectrometry (GC/MS) analysis in our laboratory.

Time-resolved CO_2, CO, NO_2, VOC, temperature, and relative humidity data were loaded to Excel spreadsheets and descriptive statistics calculated for the 8:00 a.m. to 3:00 p.m. school day period. These included peak and time-weighted-average (TWA) CO_2, CO, and NO_2 concentrations, and temperature and humidity ranges and averages. The aerosol spectrometer data were analyzed first by the instrument's specialty software, which provided estimates of the PM_{10} and $PM_{2.5}$ particle size fraction concentrations at 6-s intervals. These were then averaged over the school day to obtain TWA concentrations. Formaldehyde coupon measures were not analyzed, as discussed below.

GC/MS analysis was conducted using an Agilent 6890/5973N system (Santa Clara, CA) with Gerstel Thermal Desorption unit (Linthicum, MD). All sample tubes were reconditioned for 30 min at 300 °C with 100 mL/min ultra-high-purity nitrogen prior to field

Table 1
Elementary school characteristics and number of sampling days.

School	Original Construction Date	Major Facility Improvements	Seasons Sampled
A	2006	All new construction	Cold, Mild
B	1967	New classrooms, core facility expansion, chiller	Cold, Mild, Warm
C	1984	HVAC rehabilitation	Cold, Mild, Warm
D	2011	All new construction	Cold, Mild, Warm
E	1937	New classrooms, core facility expansion	Cold, Mild, Warm
F	1920	HVAC upgrade	Mild, Warm
G	1919	New classrooms, core facility expansion	Mild, Warm
H	1930	New classrooms, core facility expansion	Mild, Warm
I	1928	New classrooms, core facility expansion	Mild, Warm
J	1941	New classrooms, core facility expansion	Mild
K	1930	New classrooms, core facility expansion	Mild, Warm
L	1951	New classrooms, core facility expansion	Mild, Warm

deployment and were analyzed by desorbing at 280 °C for 12 min. The desorbed sample was pre-concentrated onto a Tenax packed inlet liner cooled to −30 °C and delivered to the GC column without split. GC conditions were: 1.5 mL/min column flow, oven initial temperature of 35 °C, ramped at 10 °C/min to 260 °C with 4-min hold time. MS was operated in scan mode from 30 to 350 m/z after conducting a standard tune (S-tune) to maximize spectral match of unknown compounds. The GC/MS system was calibrated in toluene equivalents from 1 to 250 ng per sample. Calibration samples were collected from a streamline injection system set to output 20 ppb and 200 ppb toluene. A sampling manifold was placed on the end of the calibration air stream and a reconditioned sorbent tube was used to collect the volume of calibration air that would yield the desired mass of toluene. All results were calibrated in toluene equivalents for approximate quantitation. Calibration correlation coefficients exceeded the analytical threshold of $R^2 \geq 0.995$.

2.2. Ventilation rate modeling

Occupant-generated CO_2 provides a convenient "natural tracer gas" for characterizing dilution ventilation. Occupants expire CO_2 at rates that depend on body size, diet, and level of activity [3]. The volume of CO_2 produced per volume of oxygen consumed is termed the "respiratory quotient", or RQ. Using equations presented in ASTM D6245-12 [3] and assuming an $RQ = 0.83$, Batterman [24] presents an expression for CO_2 volumetric generation rate G_p for school age children of

$$G_p = 1.4304\ S\ MET \tag{1}$$

where S is the Dubois surface area [25] estimate in square meters (m^2) based on height and weight, and MET is the metabolic rate per unit of surface area in Watts per square meter (W/m^2). S varies from about 0.8 to 1.4 m^2 for elementary school age children [3]. Batterman presents a table of estimates of CO_2 generation rates for adults and school age children, based on recent demographic data, in which $MET = 1.4$ for children and 1.7 for adults. For 8–10 year old children in grade level 3.5, the CO_2 generation rates for boys and girls are nearly identical at 0.216 and 0.217 L/min, respectively; the values for adult males and females are 0.500 and 0.442 L/min. These are the most current CO_2 generation estimates available.

CO_2 generation into the occupied space and its dilution via the HVAC system during a short time interval Δt can be modeled as a mass balance:

$$V\Delta C_{indoors} = (G_p + Q'C_{outdoors} - Q'C_{indoors})\Delta t, \tag{2}$$

where V is the volume of the space, $\Delta C_{indoors}$ is the incremental change in indoor concentration during the interval Δt, G_p is the volumetric rate of CO_2 generation into the space, $C_{outdoors}$ is the CO_2 concentration outdoors, and Q' is the *effective* fresh air ventilation rate in the zone of measurement. It should be emphasized that Q' is not the same as the actual flow rate at which fresh air is brought in by the HVAC system, and in fact will be substantially less than that value, because it includes the effects of imperfect mixing of incoming air with the room air [20]. When equation (2) is expressed in calculus terms, the variables may be separated and the equation integrated to yield:

$$C_{t+1} = \frac{10^6 G_p}{Q'}\left\{1 - exp\left(-\frac{Q'\Delta t}{V}\right)\right\} + (C_t - C_{outdoors})\, exp\left(-\frac{Q'\Delta t}{V}\right) + C_{outdoors} \tag{3}$$

where C_t and C_{t+1} are the CO_2 concentrations in ppm at the beginning and end of a short time interval Δt. The 10^6 value scales the dimensionless volumetric CO_2 generation rate G_p to ppm (by volume) units. Readers familiar with industrial ventilation concepts will recognize the first term of the equation as representing contaminant buildup in the space due to CO_2 generation, and the second term as representing decay in the concentration due to dilution with outdoor air containing a concentration $C_{outdoors}$ of CO_2.

Equation (3) provides a non-steady state model for estimating one of the components, given values of the other components. Thus, given values for room volume, CO_2 generation rate, and indoor and outdoor CO_2 concentrations, the effective ventilation rate Q' can be estimated over a defined time period. This is termed the Transient Mass Balance method of modeling, as described by Batterman [24,26] and also employed by Scheff et al. [20].

Our approach was similar to that of Scheff et al. [20] in that we modeled the school day in increments using a least squares solver. However, whereas they estimated fresh air ventilation rates for each hour of the school day, we modeled time periods identified as having a buildup or decay pattern. For each data set, the model was used to estimate effective ventilation rates over time periods of buildup and decay as determined from graphs of the indoor CO_2 concentration for periods of known occupancy (number of boys, girls, and teachers). Q' for the period was estimated by least squares fitting using the Excel Solver Generalized Reduced Gradient Algorithm (GRG2) utility [27]. These period estimates were then time-averaged to obtain a TWA effective ventilation rate estimate for the school day.

3. Results

3.1. Outdoor measures

3.1.1. Nitrogen dioxide

Outdoor NO_2 concentrations were rarely above 10 ppb and did not exceed 190 ppb. This high occurred during an early morning 5-min "blip" in the measures in one classroom during student drop-off, likely due to vehicle exhaust; NO_2 did not exceed 30 ppb in the other three events. This was consistent with 2015 Oklahoma City 8-h average ambient monitoring data reported by the Oklahoma Department of Environmental Quality (ODEQ), for which 1-h concentrations averaged 45 ppb (2016 data were not yet available for comparison) [28].

3.1.2. Carbon monoxide

No outdoor CO was detected, though near-road 8-h averages in the Oklahoma City have been reported to average around 2 ppm [28]. This was not surprising, since the schools were all in residential areas and outdoor monitors were placed next to the school building or on the roof. The outdoor monitors were therefore not close to major roadways.

3.1.3. Particulate matter

Due to aerosol spectrometer malfunctions during cold month measures, only one sampling event in May provided TWA outdoor PM_{10} and $PM_{2.5}$ measures during schools hours. PM_{10} values ranged from 5.5 to 173 μg/m^3 with a TWA of 30 μg/m^3, which was consistent with the Oklahoma City area 24-h average ambient PM_{10} and $PM_{2.5}$ concentrations of 24 and 12.4 μg/m^3, respectively, reported for 2015 (2016 data were not yet available for comparison) [28].

3.1.4. Volatile organic compounds

Maximum outdoor VOC concentration measured via photoionization (PID) by the air quality monitor was 80 ppb isobutylene equivalents. School hours TWA VOC by PID ranged from 0.1 to 52.2 ppb.

3.1.5. Formaldehyde

Half-hour average colorimetric formaldehyde readings were below the instrument's Limit of Detection (LOD) of approximately 10 ppb for all outdoor measures during school hours. Formaldehyde is not normally encountered in ambient air, and outdoor TWA school hour VOC (which includes formaldehyde) concentrations measured by PID and GC/MS were below 80 ppb for all events, so this result is plausible.

3.1.6. Carbon dioxide

Maximum outdoor CO_2 concentrations ranged from 231 to 676 ppm during school hours, with TWA concentrations below 300 ppm during the three measurement events in February but 477 ppm during the event in May. CO_2 levels were mostly on the lower end of the range expected for urban environments.

3.2. Indoor measures

3.2.1. Temperature and relative humidity

There were 58 indoor measurement data sets in total. Time-weighted average temperatures for the 8:00 a.m. to 3:00 p.m. school day period were generally within or close to the ranges recommended by the American Society of Heating, Refrigerating, and Air-Conditioning Engineers (ASHRAE), which are 20–24 °C (68–75 °F) in winter and 23–26 °C (73–79 °F) in summer [29]. The within-day span of temperatures rarely exceeded 6 °C (10.8 °F). Only school I was within the recommended ranges for all visits. Classrooms outside the range in the cold season were slightly too warm. None of the schools appeared to be consistently too warm. Maximum TWA temperatures were in the 28–29 °C range (82–84 °F).

3.2.2. Nitrogen dioxide

NO_2 data collected after mid-May were deemed unreliable due to problems with the IAQ meter NO_2 sensors. Earlier indoor measures rarely detected NO_2 during school hours, and all of the school hours TWA measures were near 0 ppb NO_2.

3.2.3. Carbon monoxide

Indoor CO was detected only sporadically or not at all in most sampling events. TWA CO concentrations were 0 ppm for 42 of the 58 indoor IAQ meter samples. The highest TWA CO concentrations were 10.0, 7.5, and 2.9 ppm and appeared to be due to CO spikes during student drop-off and pickup times. Nearby classrooms sampled at the same time had little or no CO measured, suggesting localized penetration of vehicle exhaust into the affected classrooms. An interesting observation outside of school hours in one school was a sudden and recurring peak CO concentration in late evening that reached over 60 ppm but resolved within about 1 h. It was determined that an indoor propane-fired floor polisher was being used during those periods (since discontinued).

3.2.4. Particulate matter

Indoor $PM_{2.5}$ and PM_{10} TWA concentrations ranged from 1.5 to 98.3 μg/m^3 for $PM_{2.5}$ (after excluding one $PM_{2.5}$ outlier) and from 0.5 to 84.3 μg/m^3 for PM_{10}. The values for each were approximately log-normally distributed with geometric means of 13.5 and 18.7 μg/m^3, respectively, with similar geometric means of 3.1 and 3.3 (after excluding the $PM_{2.5}$ outlier).

3.2.5. Volatile organic compounds

Indoor TWA VOC concentrations measured by PID ranged from 5 to 932 ppb isobutylene equivalents. Sorbent tube 12-h TWA VOC analyzed by GC/MS for five of the schools during the first round of sampling in February–March ranged from 0.1 to 30 ppb toluene equivalents as shown in Table 2. VOC species typically found included those associated with fragrances (limonene, pinene, hexanal, octanal) and solvents (toluene, xylenes, glycol ethers). Since the GC/MS was calibrated with a lower limit of quantitation of 1 ng, the 12-h TWA VOC measures are left-censored with any chemical species below the 1 ng threshold not contributing to the total mass. This is different from the operation principle of the PID VOC monitor, which measures all chemical species simultaneously without censoring individual compounds at low levels. This could explain the apparent difference between the PID and sorbent estimates of total VOC.

Indoor VOC levels were substantially higher than outdoor VOC levels as might be expected due to the indoor accumulation of exhaled VOC and VOC from dry markers, cleaning products, and other indoor sources. These concentrations were in the range reported for schools by Brown et al. [30].

Table 2
Total VOC as measured by GC/MS for five of the schools.

School	VOC Concentration (ppb toluene equivalents)		
	Lowest	Median	Highest
A	0.2	2.0	3.0
B	0.1	2.0	30.0
C	1.0	7.0	12.0
D	0.1	2.0	9.0
E	0.1	5.0	12.0

3.2.6. Formaldehyde

Formaldehyde was detected above the colorimetric method LOD during school hours only in school L, and only in the mild season visit (event 1). Each of two classrooms registered measurable formaldehyde throughout the school day and also overnight, with TWA concentrations of 15 ppb in both rooms and peaks of 17 and 18 ppb. The LOD was approximately 11 ppb. It is unclear what source might account for this measure.

3.2.7. Carbon dioxide

Class sizes were normally distributed and ranged from 16 to 33 students with a mean of 23.9 (standard deviation 4.01). Room volume per student ranged from 6.6 to 14.2 m^3/student (232–503 ft^3/student) with a mean of 9.0 and median of 8.6 m^3/student (318and 305 ft^3/student). Peak and TWA CO_2 concentrations by school and sampling event are shown in Table 3. Peak values ranged from 453 ppm in school F in the mild season (event 1) to 4751 ppm in school A in the mild season (event 2), and TWA values from 411 ppm in school F in the mild season (event 1) to 2046 ppm in school L in the mild season (event 1).

The cumulative distributions of time below various CO_2 concentrations for the full school day and for the morning and afternoon sessions separately are also shown in Table 3. If "very well ventilated" is defined for our purposes as perhaps 90% or more of the time < 1000 ppm CO_2, then only schools A, I, and K generally met the criterion. Similarly, if "reasonably well ventilated" is defined as perhaps 90% or more of the time < 1500 ppm CO_2, then schools B, C, and J generally met this criterion. School L was the most consistently poorly ventilated, with CO_2 concentrations above 1500 ppm an average of 56% of the time, and especially in the afternoons with concentrations > 1500 ppm approximately 70% of the time and > 2000 ppm 48% of the time. The other schools showed differences in the adequacy of their fresh air ventilation across seasons and sometimes across rooms during the same sampling event, perhaps due to differences in classroom volume or number of students.

3.3. Fresh air ventilation rate modeling

Of the 58 indoor measurement data sets, 7 could not be modeled due to incomplete room dimension data and 2 could not be modeled due to anomalous CO_2 concentration patterns. The remaining 49 were individually broken down into periods of buildup and decay and each segment modeled as previously described. Measured outdoor CO_2 data were used for four of the sampling events, and the rest were modeled assuming constant outdoor CO_2 levels of 350 ppm based on our own measurement experience and typical urban CO_2 levels described in the literature. Effective ventilation rates were modeled in air change per hour (ACH) units, and these values were then converted to liters per second units for comparison to ventilation standards.

Representative CO_2 concentration patterns for well-ventilated and

Table 3
Carbon dioxide levels and distributions, by school and season.

School	Event	Season	Location	TWA CO_2 (ppm)	Peak CO_2 (ppm)	Cumulative fraction of time CO_2 < concentration - school day						Cumulative fraction of time CO_2 < concentration - morning
						< 1000 ppm	< 1500 ppm	< 2000 ppm	< 2500 ppm	< 3000 ppm	< 3500 ppm	< 1000 ppm
A	1	Cold	Outdoors	267								
			Room 1	459	572	1.00						1.00
	2	Mild	Room 1	644	4751	0.99	0.99	0.99	1.00			1.00
			Room 2	785	1018	0.99	1.00					0.99
B	1	Cold	Outdoors	288								
			Room 1	682	942	1.00						1.00
	2	Mild	Room 1	865	1184	0.76	1.00					0.97
			Room 2	838	1238	0.79	1.00					1.00
	3	Warm	Room 1	961	1333	0.47	1.00					0.54
			Room 2	1129	1645	0.27	0.93	1.00				0.44
C	1	Cold	Outdoors	205								
			Room 1	1169	1635	0.26	0.90	1.00				0.45
	2	Mild	Room 1	994	1475	0.62	1.00					0.55
			Outdoors	477								
	3	Warm	Room 2	732	1202	0.99	1.00					1.00
D	1	Cold	Room 1	684	845	1.00						1.00
			Room 2	724	985	1.00						1.00
	2	Mild	Room 1	827	1054	0.87	1.00					0.87
			Room 2	940	1157	0.47	1.00					0.28
	3	Warm	Room 1	1489	2052	0.10	0.47	0.98	1.00			0.18
			Room 2	1595	2296	0.10	0.36	0.81	1.00			0.18
E	1	Cold	Room 1	1325	2036	0.19	0.63	0.99	1.00			0.32
			Room 3	904	1572	0.64	0.98	1.00				0.94
	2	Mild	Room 2	1132	2069	0.48	0.75	1.00				0.37
			Room 3	1266	2339	0.39	0.67	0.88	1.00			0.68
	3	Warm	Room 1	1500	3485	0.40	0.61	0.67	0.78	0.91	1.00	0.67
			Room 3	1540	2678	0.18	0.43	0.83	0.95	1.00		0.27
F	1	Mild	Room 1	411	453	1.00						1.00
			Room 3	1495	2278	0.02	0.53	0.91	1.00			0.04
	2	Mild	Room 2	955	1173	0.74	1.00					0.67
			Room 3	1385	1815	0.10	0.57	1.00				0.17
	3	Warm	Room 2	826	1048	0.94	1.00					0.96
			Room 9	598	961	1.00						1.00
G	1	Mild	Room 1	1172	1668	0.27	0.88	1.00				0.42
			Room 2	1093	1525	0.27	1.00					0.33
	2	Mild	Room 1	1317	2047	0.23	0.70	0.99	1.00			0.28
			Room 2	1360	1794	0.10	0.68	1.00				0.17
	3	Warm	Room 2	833	1110	0.86	1.00					0.76

(*continued on next page*)

Table 3 (*continued*)

School	Event	Season	Location	TWA CO_2 (ppm)	Peak CO_2 (ppm)	Cumulative fraction of time CO_2 < concentration - school day						Cumulative fraction of time CO_2 < concentration - morning
						< 1000 ppm	< 1500 ppm	< 2000 ppm	< 2500 ppm	< 3000 ppm	< 3500 ppm	< 1000 ppm
H	1	Mild	Room 1	1191	1540	0.16	0.94	1.00				0.22
			Room 2	1169	1603	0.16	0.98	1.00				0.19
	2	Warm	Room 2	1030	1436	0.40	1.00					0.50
			Room 3	1724	2511	0.10	0.28	0.65	0.99	1.00		0.17
	3	Warm	Room 2	1321	1626	0.08	0.76	1.00				0.15
			Room 3	1742	2624	0.09	0.28	0.68	0.94	1.00		0.16
I	1	Mild	Room 1	615	858	1.00						1.00
			Room 2	563	796	1.00						1.00
	2	Warm	Room 1	658	951	1.00						1.00
			Room 2	658	832	1.00						1.00
	3	Warm	Room 1	736	1105	0.96	1.00					0.93
			Room 2	574	897	1.00						1.00
J	1	Mild	Room 1	1064	1587	0.42	0.91	1.00				0.56
			Room 2	904	1342	0.60	1.00					0.65
K	1	Mild	Room 1	578	839	1.00						1.00
			Room 2	633	1048	0.98	1.00					0.97
	2	Warm	Room 1	653	980	1.00						1.00
			Room 2	516	590	1.00						1.00
	3	Warm	Room 1	1329	2186	0.22	0.71	0.87	1.00			0.32
L	1	Mild	Room 1	2046	2744	0.07	0.13	0.36	0.87	1.00		0.12
			Room 2	1781	2985	0.22	0.31	0.54	0.89	1.00		0.39
	2	Warm	Room 1	1351	2154	0.23	0.59	0.90	1.00			0.32
			Room 2	1358	2552	0.26	0.60	0.80	0.98	1.00		0.48
	3	Warm	Room 1	1401	2255	0.22	0.55	0.88	1.00			0.23

School	Cumulative fraction of time CO_2 < concentration - morning					Cumulative fraction of time CO_2 < concentration - afternoon					
	< 1500 ppm	< 2000 ppm	< 2500 ppm	< 3000 ppm	< 3500 ppm	< 1000 ppm	< 1500 ppm	< 2000 ppm	< 2500 ppm	< 3000 ppm	< 3500 ppm
A											
						1.00					
						0.99	0.99	0.99	0.99	0.99	0.99
	1.00					1.00					
B											
						1.00					
	1.00					0.48	1.00				
						0.52	1.00				
	1.00					0.38	1.00				
	1.00					0.05	0.85	1.00			

(*continued on next page*)

Table 3 (*continued*)

School	Cumulative fraction of time CO_2 < concentration - morning					Cumulative fraction of time CO_2 < concentration - afternoon					
	< 1500 ppm	< 2000 ppm	< 2500 ppm	< 3000 ppm	< 3500 ppm	< 1000 ppm	< 1500 ppm	< 2000 ppm	< 2500 ppm	< 3000 ppm	< 3500 ppm
C											
	1.00					0.00	0.77	1.00			
	1.00					0.71	1.00				
						0.98	1.00				
D						1.00					
						1.00					
	1.00					0.88	1.00				
	1.00					0.72	1.00				
	0.63	0.98	1.00			0.00	0.26	0.98	1.00		
	0.60	0.95	1.00			0.00	0.03	0.62	1.00		
E	0.58	0.98	1.00			0.00	0.71	1.00			
	1.00					0.34	0.97	1.00			
	0.65	1.00				0.63	0.87	1.00			
	0.89	0.97	1.00			0.00	0.37	0.77	1.00		
	0.96	1.00				0.05	0.13	0.22	0.49	0.79	1.00
	0.51	0.75	0.92	1.00		0.05	0.33	0.94	1.00		
F						1.00					
	0.75	1.00				0.00	0.24	0.80	1.00		
	1.00					0.84	1.00				
	0.90	1.00				0.00	0.13	1.00			
	1.00					0.91	1.00				
						1.00					
G	0.98	1.00				0.07	0.76	1.00			
	1.00					0.18	0.99	1.00			
	0.94	1.00				0.16	0.37	0.98	1.00		
	0.64	1.00				0.00	0.73	1.00			
	1.00					0.99	1.00				
H	0.95	1.00				0.09	0.92	1.00			
	0.96	1.00				0.12	1.00				
	1.00					0.27	1.00				
	0.33	0.67	1.00			0.00	0.22	0.62	0.98	1.00	
	0.78	1.00				0.00	0.73	1.00			
	0.38	0.88	1.00			0.00	0.14	0.42	0.87	1.00	
I						1.00					
						1.00					
						1.00					
						1.00					
	1.00					1.00					
						1.00	-	-	-	-	-
J	1.00					0.23	0.79	1.00			
	1.00					0.54	1.00				

(*continued on next page*)

Table 3 (*continued*)

School	Cumulative fraction of time CO_2 < concentration - morning					Cumulative fraction of time CO_2 < concentration - afternoon					
	< 1500 ppm	< 2000 ppm	< 2500 ppm	< 3000 ppm	< 3500 ppm	< 1000 ppm	< 1500 ppm	< 2000 ppm	< 2500 ppm	< 3000 ppm	< 3500 ppm
K						1.00					
	1.00					1.00					
						1.00					
						1.00					
	0.94	1.00				0.09	0.41	0.70	1.00		
L	0.22	0.57	0.85	1.00		0.00	0.00	0.08	0.90	1.00	
	0.55	0.77	1.00			0.00	0.00	0.23	0.74	1.00	
	0.62	1.00				0.13	0.54	0.79	1.00		
	0.79	1.00				0.00	0.37	0.54	0.96	1.00	
	0.48	0.81	1.00			0.21	0.63	0.96	1.00		

poorly-ventilated classrooms, and associated modeling results, are shown in Fig. 1. The solid lines are the measured values and the dotted lines represent the values predicted by the modeling. The room in panel A was very well ventilated as demonstrated by the generally low CO_2 levels and rapid decay during non-occupancy periods (recess, lunch, and after 3:00 p.m.). In contrast, the room in panel B was poorly ventilated, with high CO_2 levels and a much slower decay rate.

There was considerable variability in the time-concentration patterns, which would be expected given differences in occupancy patterns and HVAC systems. Fig. 2 shows the influence of a rapidly cycling HVAC system during warm weather in school G, providing what appears to be an increased fresh air flow with each cycle.

Estimated effective fresh air ventilation rates are shown in Table 4 in both ACH and L/sec units. ASHRAE-recommended classroom ventilation rates in L/sec are also shown, for combined people-and-space related pollutants (5 × number of persons + 0.6 × area in square meters) and people-related pollutants only (5 L/sec per person) [31].

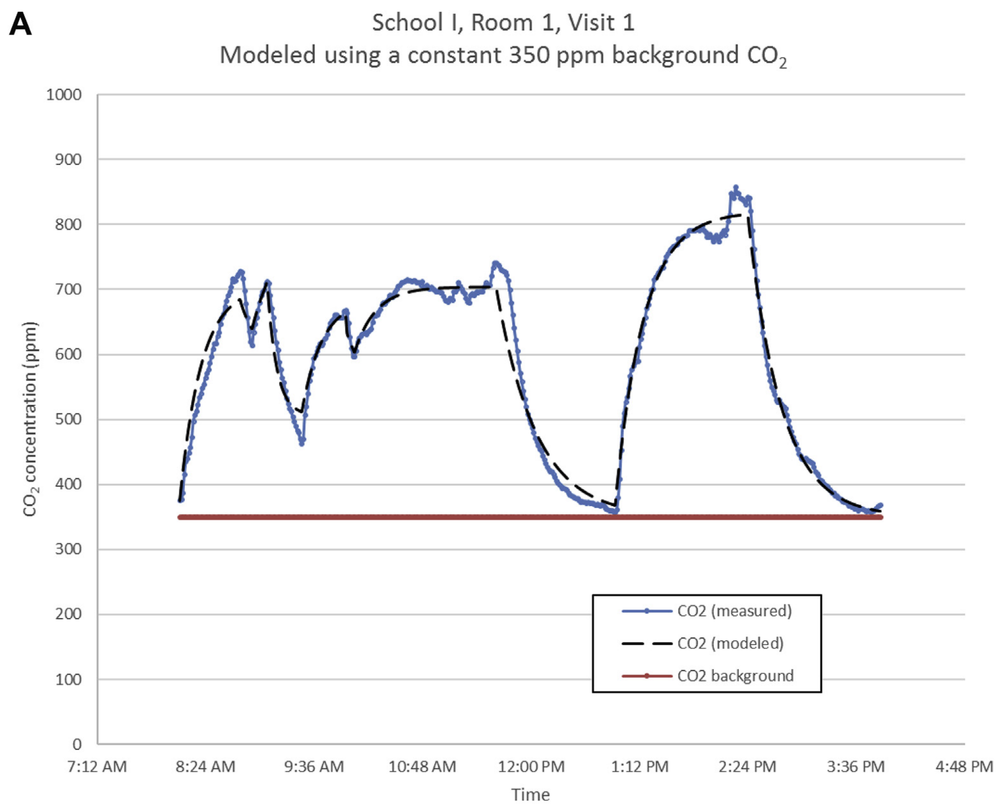
The blocks with bold text in the *TWA CO_2 (ppm)* column are those for which the TWA concentration was below 1000 ppm. The blocks with bold text in the *Q′ (L/sec) (modeled with 350 ppm outdoor CO_2)* column are those events for which the effective fresh air ventilation rate, as modeled for 350 ppm background, was at least 90% of that recommended by ASHRAE for control of people-related pollutants (i.e. 5 L/s per person, or 10.6 ft^3/min [cfm] per person). There is a high correspondence between *Q′* estimates above 90% of the ASHRAE-recommended effective ventilation rate for people-related pollutant control and maintenance of TWA CO_2 concentrations below 1000 ppm. This suggests that the modeling technique may be providing reasonably accurate estimates of effective fresh air ventilation rates for control of people-related pollutants, in agreement with the results of Batterman et al. [26]. The classrooms evaluated had no copiers, printers, or other office machines that might contribute pollutants, and VOC measures did not indicate any notable levels of VOC that might have originated from building materials or cleaning products, so the technique may be especially appropriate for these and similar classrooms.

4. Discussion

The school-day CO, NO_2, VOC, and formaldehyde indoor pollutant levels in the schools were extremely low to non-detectable. Penetrations of vehicle-related pollutants to classrooms from outdoors occurred only occasionally and were brief and of minimal concentration. Temperature and relative humidity were well controlled for the most part, though temperatures exceeded ASHRAE-recommended levels slightly in some cases. Only CO_2 concentrations were found to exceed desirable levels in a number of schools.

CO_2 is no longer considered an innocuous indicator of other people-related pollutants; rather, it is now believed to be responsible for occupant discomfort, performance decrement, and health effects, at least for adults [4,5]. TWA CO_2 levels exceeded 1000 ppm in 28 of 58 sampling events, and peak CO_2 levels exceeded 1000 ppm in 44 of 58 events. This was consistent with observations in previous studies that U.S. schools are often inadequately ventilated [6,11,13,32]. There was no clear correspondence between the adequacy of ventilation and either the school's original build date or the major facility improvements achieved during renovation (Table 1). Indeed, two of the "very well ventilated" schools that maintained CO_2 levels below 1000 ppm greater than 90% of the time were also two of the oldest schools (School I, built in 1928, and School K, built in 1930). The observed differences in the CO_2 buildup and decay patterns over the school day, as well as the differences in HVAC performance across seasons for a given school in many cases, suggest HVAC system type or operating conditions play a major if not dominant role in CO_2 control. Unfortunately, detailed HVAC system data were not available for study.

The high CO_2 levels have important implications for the academic achievement of students in these classrooms. Inadequate fresh air



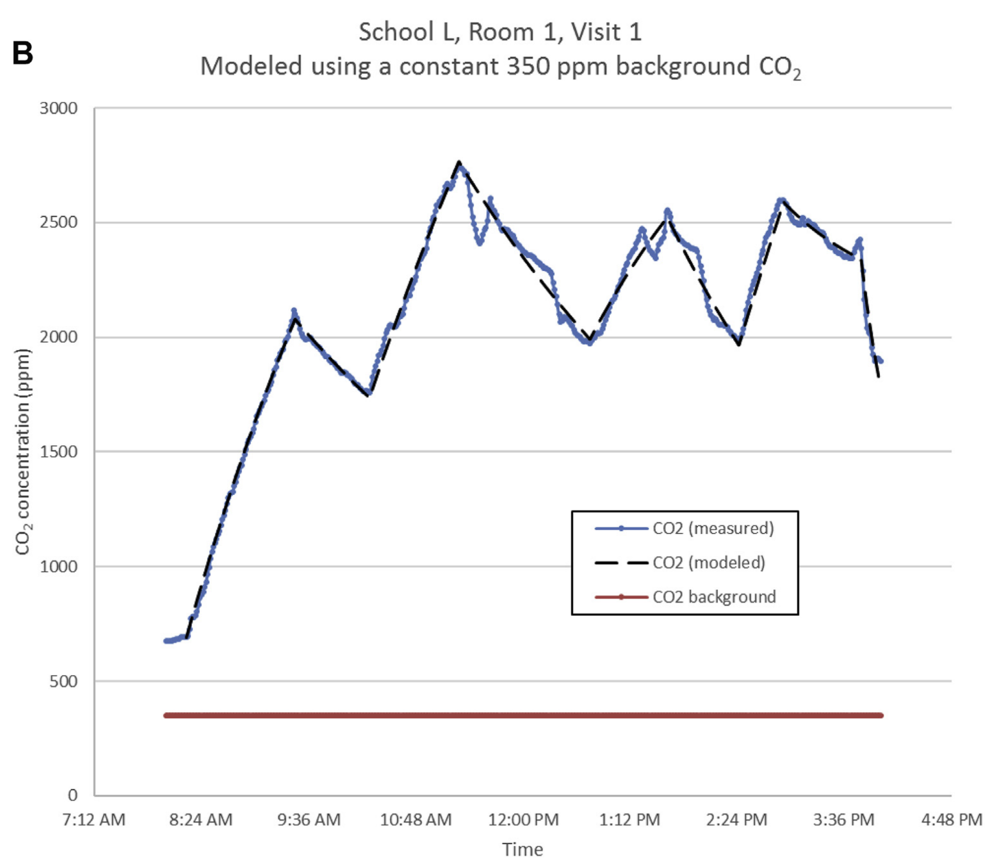


Fig. 1. Representative plot of CO_2 concentration measurements over the course of a school day for a well-ventilated classroom (panel A) and a poorly ventilated classroom (panel B). Large concentration dips correspond to periods when the students and the teacher were not in the classroom, such as during recess, the lunch period, and after 3:00. The dotted line represents predicted values from the fitted model.

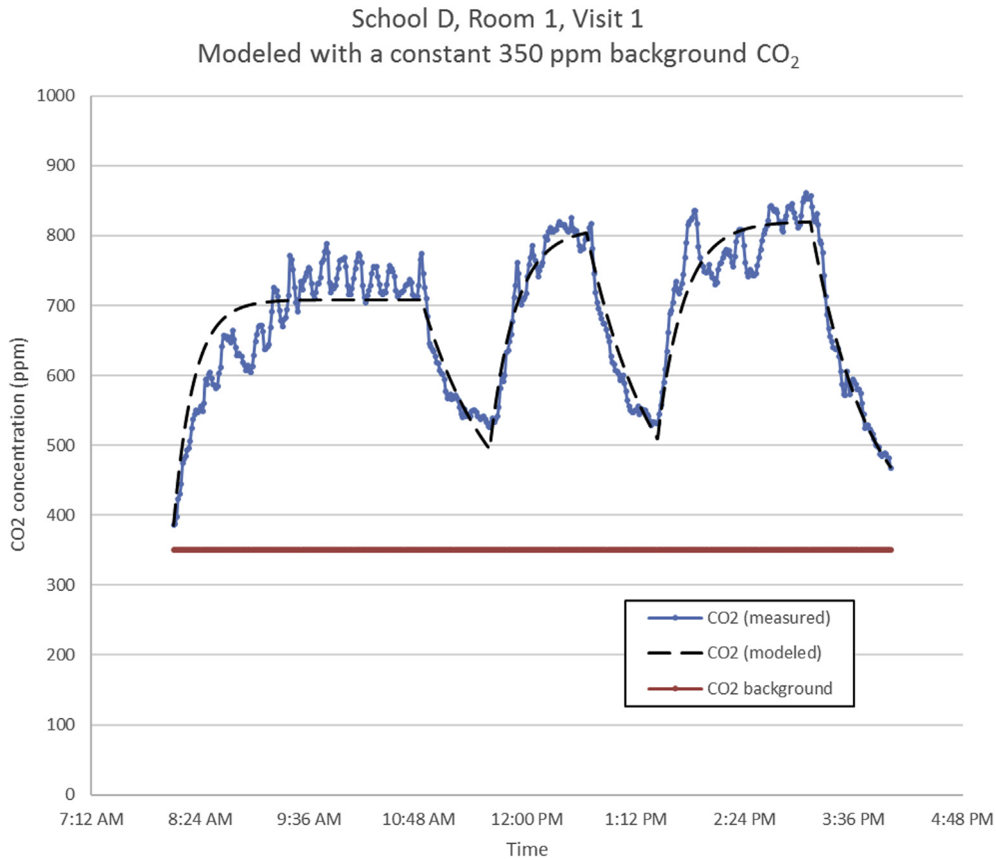


Fig. 2. CO2 concentration plot suggesting cyclic operation of the HVAC system and fresh air supply. The effective ventilation was very good, maintaining CO_2 concentrations well below 1000 ppm.

ventilation in classrooms, as reflected in high CO_2 levels, has been shown to adversely affect student performance on both numerical and language-based tasks [6,7,12,33–36]. Wargocki and Wyon [34–36] demonstrated task performance improvements (measured by speed) of up to 14% with a doubling of fresh air supply rate over the range of 3 to 9.5 L/sec per student. Haverinen-Shaughnessy and colleagues saw approximately linear performance increases as ventilation was increased over the range of 0.9 to 7 L/sec per student [6,7]. The slope of this approximately linear relationship of decreasing performance with decreasing ventilation rate would be the same whether ventilation is measured as total fresh air ventilation rate or effective ventilation rate.

The correspondence between estimated effective fresh air ventilation rates as modeled using the transient mass balance approach, the ASHRAE-recommended rate for people-related pollutants, and control of CO_2 levels was good. The modeled rates were similar in range to those estimated by Scheff et al. [20] for a mechanically ventilated classroom. The effective fresh air ventilation rate in each classroom was modeled assuming an idealized single zone system with uniform instantaneous mixing and no fresh air infusion by any route but the HVAC system. The HVAC system will certainly not have perfect mixing, as demonstrated by Scheff et al. [20] when they compared modeled fresh air ventilation rates with rates actually measured at the HVAC intakes. Their work indicated mixing factors, as the ratio of modeled to measured rates, to average 0.40 for a classroom and 0.61 for a cafeteria. Such ratios could not be estimated in this work due to lack of information on the actual fresh air supply rates, but are only of minor interest. The important value is the effective ventilation rate, Q', as modeled, which includes the effects of imperfect mixing.

The transient mass balance approach to effective ventilation rate modeling as used in this work is, in our view, superior to the simple steady-state, buildup, or decay methods [24,37]. The steady-state method requires that the CO_2 concentration reach equilibrium during the modeling period. Fig. 1 and the other plots of our data indicated that equilibrium was seldom achieved. The buildup method models CO_2 concentration buildup only from the time of initial occupancy assuming a constant generation rate, which is represented by the first term in equation (3). The decay method models CO_2 concentration decay only from the time a space is vacated and assumes a zero generation rate, which is represented by the second term in equation (3). Over a school day, the buildup approach can be applied at the start of the day, after recess periods, and after the lunch break, while the decay approach can be applied during recess periods and the lunch break and at the end of the day. Buildup and decay are just special cases of the general relationship between CO_2 generation rate, outdoor (dilution) air supply rate and CO_2 concentration, and room size, i.e. equation (3). By utilizing equation (3) and segmenting the day into buildup and concentration periods as reflected in the time-concentration profile, with specified CO_2 generation rate during occupied periods and zero generation during vacant periods assigned to the segments, solution of all segments using Excel Solver provides simultaneous least-squares estimates of each segment's Q' that can then be time-weighted by the segment lengths. This does not require that any assumptions be made about whether or not the system has a constant effective fresh air ventilation rate over the course of the day.

The adequacy of effective fresh air ventilation rates estimated in this work was assessed by comparing the rates to the ASHRAE-recommended fresh air ventilation rate for people-related pollutants only, i.e. 5 L/s (10.6 cfm) per person. This was considered appropriate because measurements indicated a lack of building or activity-related pollutant contributions to classroom IAQ, and because there was no attempt to adjust the estimated *effective* fresh air ventilation rate in the measurement zone to a *total* fresh air ventilation rate at the air inlet

Table 4
Estimated effective fresh air ventilation rates.

School	Event	Season	Location	TWA CO_2 (ppm)	TWA ACH (modeled with actual outdoor CO_2)	TWA ACH (modeled with 350 ppm outdoor CO_2)	Q' (L/sec) (modeled with actual outdoor CO_2)	Q' (L/sec) (modeled with 350 ppm outdoor CO_2)	ASHRAE Q' (L/sec) people and area	ASHRAE Q' (L/sec) people only
A	1	Cold	Room 1	**459**	5.9	7.4	374	**470**	152	100
	2	Mild	Room 1	**644**		3.4		**216**	152	100
			Room 2	**785**		2.8		**134**	137	95
B	1	Cold	Room 1	**682**	4.2	5.1	263	**320**	198	150
	2	Mild	Room 1	**865**		2.4		**151**	198	150
			Room 2	**838**		3.3		**202**	188	140
	3	Warm	Room 1	**961**		2.2		**138**	198	150
			Room 2	1129		1.6		98	188	140
C	1	Cold	Room 1	1169	1.6	1.9	263	100	173	130
	2	Mild	Room 1	**994**	2.4	1.9	263	100	173	130
	3	Warm	Room 2	**732**		3.7		**186**	152	110
D	1	Cold	Room 1	**684**		3.4		**209**	182	135
			Room 2	**724**		2.8		**175**	183	135
	2	Mild	Room 1	**827**		2.5		**153**	182	135
			Room 2	**940**		2.2		**137**	183	135
	3	Warm	Room 1	1489		1.2		74	187	140
			Room 2	1595		0.8		50	178	130
E	1	Cold	Room 1	1325		1.2		67	130	130
			Room 3	**904**		2.7		**159**	145	145
	2	Mild	Room 2	1132	Incomplete data				145	
			Room 3	1266		2.0		118	145	145
	3	Warm	Room 1	1500		1.9		**106**	105	105
			Room 3	1540		0.8		47	125	125
F	1	Mild	Room 1	**411**	Incomplete data				170	
			Room 3	1495		1.3		85	225	170
	2	Mild	Room 2	**955**	Incomplete data				165	
			Room 3	1385		1.6		105	225	170
	3	Warm	Room 2	**826**	Incomplete data				85	
			Room 9	**598**		2.5		**164**	170	115
G	1	Mild	Room 1	1172		1.3		69	171	130
			Room 2	1093	Could not model				110	
	2	Mild	Room 1	1317		1.0		53	171	130
			Room 2	1360		0.9		46	110	110
	3	Warm	Room 2	**833**		2.1		**108**	110	110
H	1	Mild	Room 1	1191	Incomplete data				100	
			Room 2	1169		0.7		53	149	100
	2	Warm	Room 2	1030		1.1		83	149	100
			Room 3	1724		1.0		47	174	135
	3	Warm	Room 2	1321		0.4		30	149	100
			Room 3	1742		0.9		43	174	135
I	1	Mild	Room 1	**615**		4.0		**237**	160	120
			Room 2	**563**		4.2		**249**	125	125
	2	Warm	Room 1	**658**		3.1		**184**	160	120
			Room 2	**658**		3.6		**214**	125	125
	3	Warm	Room 1	**736**		3.8		**225**	160	120
			Room 2	**574**		*			125	125
j	1	Mild	Room 1	1064	Incomplete data				115	
			Room 2	**904**	Incomplete data				110	
K	1	Mild	Room 1	**578**		3.4		**186**	146	100
			Room 2	**633**		6.6		**342**	90	90
	2	Warm	Room 1	**653**		4.6		**252**	176	130
			Room 2	**516**	Could not model				125	
	3	Warm	Room 1	1329		1.5		82	176	130
L	1	Mild	Room 1	2046		0.3		18	175	125
			Room 2	1781		1.0		62	175	125
	2	Warm	Room 1	1351		0.7		43	185	135
			Room 2	1358		0.6		37	175	125
	3	Warm	Room 1	1401		0.8		49	185	135

such as specified in the standard. The two are not actually directly comparable unless the relationship between total outside air supply and the estimated effective dilution air flow (taking into account imperfect mixing) is known.

The types and operating modes of the school HVAC systems were not characterized in this work, but the time-concentration profiles suggested important differences in the systems. Fig. 2 clearly represents a system with a substantial but variable outside air supply, apparently varying as the system calls for cooling. In contrast, Fig. 1A suggests a system that has a steady and substantial outside air supply. The nearly linear CO_2 buildup and high concentrations in Fig. 1B suggest a system that provides little outside air. These operating modes are reflected in the TWA Q' estimates in Table 4 (3.4, 4.0, and 0.3 ACH, respectively).

It should be noted that the modeling assumed a constant CO_2 generation rate specified as the average expected for male and female third graders, as estimated by Batterman [24]. Clearly, activity levels and therefore CO_2 generation rates will not be constant all day, but intuition suggests that CO_2 generation should not vary widely from the average for normal (sedentary) classroom activities. It should also be noted that it was assumed that each of the classrooms functioned as a single zone, with all outside air (from whatever source) having a specified CO_2 concentration. While the classrooms were somewhat "closed" environments with closed doors during instruction and sealed windows, those on central HVAC systems would experience CO_2 in supply air that is influenced by the return air CO_2 concentrations and flow rates from all rooms on the system as well as the outdoor CO_2 concentration and flow rate. High CO_2 in return air from another room would effectively decrease the $C_t - C_{outdoors}$ term in equation (3) by increasing the apparent $C_{outdoors}$, thereby overestimating Q'. Classroom sizes and populations were similar across classrooms in the schools visited, so this may not have been a factor in any schools with multi-room HVAC zones. Additionally, it was not determined if there were any unused classrooms that may have shared the subject room's HVAC system. If so, the empty room would effectively increase the space volume, thereby underestimating Q'. Thus, either multi-room HVAC zones or unused spaces on the system could have influenced the Q' estimates, but in opposite directions.

5. Conclusions

Indoor air quality measures of CO, CO_2, NO_2, VOCs, formaldehyde, PM, temperature, and relative humidity were conducted in 12 Oklahoma City area elementary schools over the course of the school day during cold, mild, and warm season conditions. Only third grade classrooms were assessed. All of the schools had been recently renovated to various degrees or were newly constructed. Continuous monitoring CO_2 data, occupancy data, and space characteristics were used to model ventilation system effective fresh air supply using the Transient Mass Balance Method.

Results indicated the schools to have generally adequate temperature control, extremely low non-occupant related pollutants, and little to no incursion of outdoor vehicle-related pollutants. However, there was a lack of adequate fresh air ventilation in many cases, perhaps to the detriment of academic performance. Ventilation adequacy varied within many schools across seasons but with no consistent pattern, perhaps reflecting variations in class size as well as seasonal demands on the HVAC systems and/or HVAC seasonal operating mode. Additional study to explore these factors and their influence on classroom IAQ would be valuable. Transient mass balance method effective fresh air ventilation estimates near or above ASHRAE-recommended fresh air ventilation rates for people-related pollutants corresponded well with good CO_2 control in the classrooms.

Declarations of interest

None.

Conflicts of interest

The authors declare no conflicts of interest.

Acknowledgments

This work was supported by the US Environmental Protection Agency, project EPA-G2013-START-H1-R835634, "Balancing Sustainability, Clean Air, Health Learning Interiors, and Structural Safety when Designing and Building Schools". We extend our appreciation to the teachers and administrators of Oklahoma City Public Schools system for their cooperation and assistance. We also extend our appreciation to graduate students Sarah Kimball and Tien Hoang for their aid in data collection. This publication has not been formally reviewed by the EPA. The EPA does not endorse any products or commercial services mentioned in this publication.

References

[1] K.W. Tham, Indoor air quality and its effects on humans-A review of challenges and developments in the last 30 years, Energy Build. 130 (2016) 637–650 (15 October 2016).
[2] M. Marbury, J. Woods, Building-related illnesses, in: J. Samet, J. Spengler (Eds.), Indoor Air Pollution - a Health Perspective, Johns Hopkins University Press, Baltimore, MD, 1991, pp. 306–322.
[3] ASTM, Standard Guide for Using Indoor Carbon Dioxide Concentrations to Evaluate Indoor Air Quality and Ventilation, ASTM International, West Conshohocken, PA, 2012.
[4] J. Allen, P. MacNaughton, U. Satish, J. Vallarino, J. Spengler, Associations of congnitive function scores with carbon dioxide, ventilation, and volatile organic compound exposures in office workers: a controlled exposure study of Green and conventional office environments, Environ. Health Perspect. 124 (6) (2016) 8.
[5] U. Satish, M. Mendell, K. Shekhar, T. Hotchi, D. Sullivan, S. Streufert, W. Fisk, Is CO_2 an indoor Pollutant? Direct effects of low-to-moderate CO_2 concentrations on human decision-making performance, Environ. Health Perspect. 120 (12) (2012) 7.
[6] U. Haverinen-Shaughnessy, D. Moschandreas, R. Shaughnessy, Association between substandard classroom ventilation rates and students' academic achievement, Indoor Air 21 (2011) 11.
[7] U. Haverinen-Shaughnessy, R. Shaughnessy, Effects of classroom ventilation rate and temperature on students' test scores, PLoS One 10 (8) (2015) 14.
[8] D. Shendell, R. Prill, W. Fisk, M. Apte, D. Blake, D. Faulkner, Associations between classroom CO_2 concentrations and student attendance in Washington and Idaho, Indoor Air 14 (2004) 333–341.
[9] P. Wargocki, D. Wyon, The effects of outdoor air supply rate and supply air filter condition in classroom CO_2 concentrations and student attendance in Washington and Idaho, HVAC R Res. 13 (2) (2007) 165–191.
[10] J. Sundell, H. Levin, W. Nazaroff, W. Cain, W. Fisk, D. GHrimsrud, F. Gyntelberg, Y. Li, A. Persily, A. Pickering, J. Samet, J. Spengler, S. Taylor, C. Weschler, Ventilation rates and health: multidisciplinary review of the scientific literature, Indoor Air 21 (2011) 191–204.
[11] M. Mendell, E. Eliseeva, M. Davies, M. Spears, A. Lobscheid, W. Fisk, Association of classroom ventilation with reduced illness absence: a prospective study in California elementary schools, Indoor Air 6 (2013) 14.
[12] Z. Bako-Biro, D. Clements-Croome, N. Kochhar, H. Awbi, M. Williams, Ventilation rates in schools and pupils' performance, Build. Environ. 48 (2012) 215–223.
[13] M. Mendell, E. Eliseeva, M. Davies, A. Lobscheid, Do classroom ventilation rates in California elementary schools influence standardized test scores? Results from a prospective study, Indoor Air 26 (2016) 12.
[14] M. Mendell, G. Heath, Do indoor pollutants and thermal conditions in schools influence student performance? A critical review of the literature, Indoor Air 15 (2005) 6.
[15] L. Chatzikiakou, D. Mumovic, A. Summerfield, S. Hong, H. Altamirano-Medina, A Victorial school and a low carbon designed school: comparison of indoor air quality, energy performance, and student health, Indoor Built Environ. 23 (3) (2014) 16.
[16] A. Mumovic, J. Palmer, M. Davies, M. Orme, I. Ridley, T. Oreszczyn, C. Judd, R. Critchlow, H. Medina, G. Pilmoor, C. Pearson, P. Way, Winter indoor air quality, thermal comfort and acoustic performance of newly built secondary schools in England, Build. Environ. 44 (2009) 12.
[17] A. Lugg, W. Batty, Air quality and ventilation rates in school classrooms I: air quality monitoring, Proceedings of the CIBSE A: Build. Serv. Eng. Technol. 20 (1) (1999) 9.
[18] A. Beisteiner, D. Coley, Carbon dioxide levels and summertime ventilation rates in UK schools, Int. J. Vent. 1 (3) (2003) 7.
[19] D. Coley, A. Beisteiner, Carbon dioxide levels and ventilation rates in schools, Int. J. Vent. 1 (1) (2002) 8.
[20] P. Scheff, V. Paulius, S. Huang, L. Conroy, Indoor air quality in a middle school, Part I: use of CO_2 as a tracer for effective ventilation, Appl. Occup. Environ. Hyg 15 (11) (2000) 824–834.
[21] V. Turanjanin, B. Vucicevic, M. Jovanovic, N. Mirkov, I. Lazovic, Indoor CO_2 measurements in Serbian schools and ventilation rate calculation, Inside Energy 77

(2014) 290–296.
[22] A. Myhrvold, E. Olsen, O. Lauridsen, Indoor Environment in Schools - Pupils Health and Performance in Regard to CO2 Concentrations, Indoor Air '96: The 7th International Conference on Indoor Air Quality and Climate, Nagoya, Japan (1996), pp. 369–374.
[23] J. Gao, P. Wargocki, Y. Wang, Ventilation system type, classroom environmental quality and pupils' perceptions and symptoms, Build. Environ. 75 (2014) 12.
[24] S. Batterman, Review and extension of CO_2-based methods to determine ventilation rates with application to school classrooms, Int. J. Environ. Res. Publ. Health 14 (2017) 22.
[25] D. DuBois, E. DuBois, A formula to estimate approximate surface area, if height and weight are known, Arch. Intern. Med. 17 (1916) 9.
[26] S. Batterman, F.C. Su, A. Wald, F. Watkins, C. Godwin, G. Thun, Ventilation rates in recently constructed U.S. school classrooms, Indoor Air 27 (5) (2017) 880–890.
[27] D. Flystra, L. Lasdon, J. Watson, A. Waren, Design and use of the microsoft Excel solver, Interfaces 28 (5) (1998) 27–31.
[28] ODEQ, Air Data Report 2015, (2016) (Accessed March 23, 2017 2017).
[29] ASHRAE, Thermal Environmental Conditions for Human Occupancy, American Society for Heating, Refrigerating, and Air Conditioning Engineers, Atlanta, GA, 1992.
[30] S. Brown, M. Sim, M. Abramson, C. Gray, Concentrations of volatile organic compounds in indoor air - a review, Indoor Air 4 (1994) 12.
[31] ANSI/ASHRAE, Ventilation for Acceptable Indoor Air Quality, American Society of Heating, Refrigerating, and Air Conditioning Engineers (ASHRAE), Atlanta, GA, 2016, p. 54.
[32] J. Daisey, W. Angell, M. Apte, Indoor air quality, ventilation and health symptoms in schools: an analysis of existing information, Indoor Air 13 (2003) 12.
[33] S. Petersen, K.L. Jensen, A.L. Pedersen, H.S. Rasmussen, The effect of increased classroom ventilation rate indicated by reduced CO2 concentration on the performance of schoolwork by children, Indoor Air 26 (3) (2016) 366–379.
[34] P. Wargocki, D. Wyon, The effects of outdoor air supply rate and supply air filter condition in classrooms on the performance of schoolwork by children: ASHRAE 1257-RP, HVAC R Res. 132 (2007) 26.
[35] P. Wargocki, D. Wyon, The effects of moderately raised classroom temperatures and classroom ventilation rate on the performance of schoolwork by children: ASHRAE 1257-RP, HVAC R Res. 132 (2007) 27.
[36] P. Wargocki, D.P. Wyon, Research report on - effects of HVAC on student performance, ASHRAE J. 48 (10) (2006) 22-+.
[37] A. Persily, Evaluating building IAQ and ventilation with indoor carbon dioxide, Build. Eng. 103 (1997) 12.